

Symmetric Robust Auction Inference under Latent Information: A Note

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Abstract

This note studies symmetric auction environments with latent information. Syrgkanis et al. (2021) show that unrestricted BCE provides a foundation for robust auction inference under weak assumptions on information. This note asks whether, within a maintained symmetric auction class, the distributions generated by symmetric Bayes Nash equilibrium over admissible latent-information expansions admit an exact lower-dimensional symmetric representation. For a given payoff-relevant auction environment (base game), I define a symmetric reduced recommendation rule over bids, conditional on a bidder’s retained value, an anonymous summary of opponents’ retained values, and the public state. The preliminary theorem-level result is that this reduced object induces exactly the same distributions over bids, retained values, and public states as symmetric Bayes Nash equilibrium over admissible expansions in the maintained class. The comparison with unrestricted BCE is mainly computational and representational. Unrestricted BCE yields a linear feasibility problem over a full joint distribution. The primitive symmetric formulation works with a smaller recommendation rule, but the induced joint distribution is generated through conditional-independence products, so obedience constraints become polynomial. Thus the relevant tradeoff is not simply linear versus nonlinear, but full-joint unrestricted search versus lower-dimensional structured search.

1 Introduction

Auction outcomes depend critically on bidders’ information about payoff-relevant uncertainty. In many empirical environments, however, the analyst does not observe bidders’ information sets and may be unwilling to impose a fully specified information structure. A prominent response to this problem is to use Bayes correlated equilibrium (BCE) as an

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equilibrium-based description of outcomes under latent information. In particular, Syrgkanis et al. (2021) show that unrestricted BCE provides a useful foundation for robust auction inference under weak assumptions on information.

This note asks how that robust-inference logic changes when the analyst maintains primitive symmetry restrictions on the latent-information environment. The maintained class considered here is symmetric: auction payoffs are symmetric, retained payoff-relevant components are exchangeably distributed, and the latent additional-information layer is symmetric and conditionally independent across bidders given the retained payoff-relevant state. These restrictions are imposed on the information environment before the model is reduced to distributions over bids and payoff-relevant primitives.

The question is whether, within this maintained symmetric auction class, the outcome distributions generated by symmetric Bayes Nash equilibrium over admissible latent-information expansions admit an exact lower-dimensional symmetric representation. The theorem-level answer proposed in this note is yes. For a given payoff-relevant auction environment, the set of distributions over bids, retained values, and public states induced by symmetric Bayes Nash equilibrium, as the admissible additional-information layer ranges over the maintained class, coincides with the set induced by a symmetric recommendation rule over bids. This rule conditions only on a bidder’s own retained value, an anonymous summary of opponents’ retained values, and the public state.

This is the auction-facing analogue of the exact SBCE reduction in Lu (2026). The point is not that unrestricted BCE is replaced by a general-purpose alternative. Rather, once the analyst maintains the symmetric latent-information class, the appropriate reduced object is the one generated by that class. The reduced object carries the restrictions on symmetric behavior and on the latent additional-information layer that are not encoded by an unrestricted joint distribution over bids and payoff-relevant primitives alone.

The main difference from unrestricted BCE arises at the level of characterization. Under unrestricted BCE, one searches directly over a full joint distribution on bids and payoff-relevant primitives, so Bayes consistency and obedience are linear in the primitive decision variables. Under the primitive symmetric formulation, the primitive object is a lower-dimensional single-bidder recommendation rule, and the joint bid distribution is generated from this rule through conditional-independence products. As a result, the natural fea-

sibility problem is polynomial rather than linear. The relevant tradeoff is therefore not simply linear versus nonlinear, but full-joint unrestricted search versus lower-dimensional structured search.

The purpose of the note is correspondingly limited. It formulates the maintained symmetric auction class using auction notation, states a preliminary theorem-level equivalence result identifying the exact reduced object, and clarifies why the linear-programming characterization used for unrestricted BCE does not automatically carry over to the primitive symmetric formulation. The note does not provide a finished econometric implementation or a final sharp computational characterization.

The remainder of the note is organized as follows. Section 2 introduces a symmetric auction environment with latent information. Section 3 defines the reduced symmetric object and states a preliminary equivalence theorem. Section 4 discusses the proof logic and its relation to the main SBCE theorem. Section 5 explains why the unrestricted BCE linear-programming characterization does not automatically carry over and describes the resulting primitive-search formulation. Section 6 concludes with scope conditions and open questions.

2 A symmetric auction environment with latent information

This section introduces the maintained class of symmetric auction environments studied in the note. The setup follows the informational decomposition of Lu (2026), but uses notation chosen to reduce frictions for auction readers. I write bidder i 's action as a bid $b_i \in B$ and the retained value component as $v_i \in V$. Relative to Lu (2026), the object written there as t_i is written here as v_i , while the latent additional-information component continues to be denoted by s_i . The public state $\theta \in \Theta$ is publicly observed and common knowledge throughout; in auction applications, it may represent a common-value component or another publicly observed payoff-relevant auction state.

2.1 Bidders, bids, states, and information

There are N bidders, a finite public-state space Θ , a common finite bid space B , and a common finite retained-value space V . Let B^N denote the set of bid profiles. Bidder i

privately observes

$$(v_i, s_i) \in V \times S_i,$$

where $v_i \in V$ is the retained value component, interpreted more generally as the bidder's retained auction type, and S_i is a measurable additional-information space. The variable $s_i \in S_i$ is an additional latent-information component. Let

$$v = (v_1, \dots, v_N) \in V^N, \quad s = (s_1, \dots, s_N) \in S := \prod_{i=1}^N S_i$$

denote the retained-value profile and the latent-information profile, respectively.

Payoffs are given by

$$u_i : B^N \times V^N \times \Theta \rightarrow \mathbb{R},$$

so bidder i 's payoff may depend on the bid profile, the retained-value profile, and the public state. The common-prior information structure is

$$I = (V, (S_i)_{i=1}^N, \pi), \quad \pi \in \Delta(V^N \times S \times \Theta).$$

Let

$$\pi^{V\Theta} \in \Delta(V^N \times \Theta)$$

denote the marginal distribution of (v, θ) induced by π .

2.2 Symmetry

For any permutation φ of $\{1, \dots, N\}$, define

$$\varphi(b) := (b_{\varphi(1)}, \dots, b_{\varphi(N)}) \quad \text{and} \quad \varphi(v) := (v_{\varphi(1)}, \dots, v_{\varphi(N)}).$$

The auction environment is symmetric if, for every permutation φ , every bidder i , every bid profile $b \in B^N$, every retained value profile $v \in V^N$, and every public state $\theta \in \Theta$,

$$u_i(b, v, \theta) = u_{\varphi(i)}(\varphi(b), \varphi(v), \theta).$$

2.3 Exchangeable retained values and symmetric conditionally independent additional information

The retained value component is *exchangeable* if

$$\pi^{V\Theta}(\varphi(v), \theta) = \pi^{V\Theta}(v, \theta) \quad \text{for all } (v, \theta) \in V^N \times \Theta$$

and every permutation φ of $\{1, \dots, N\}$.

The conditional law of the additional-information profile given (v, θ) is used only on the support of $\pi^{V\Theta}$. Since $V^N \times \Theta$ is finite, for any (v, θ) with $\pi^{V\Theta}(v, \theta) > 0$, write $\pi(\cdot \mid v, \theta)$ for the conditional law of s given (v, θ) . No restriction is imposed on conditional versions off this support.

For each bidder i , permutations of bidder i 's opponents induce corresponding relabelings of the coordinates of S_i , and hence of measurable subsets of S_i . The additional-information layer is *symmetric conditionally independent* if there exists a common stochastic kernel

$$\rho(\cdot \mid v_i, v_{-i}, \theta) \in \Delta(S_i)$$

such that, for every $(v, \theta) \in V^N \times \Theta$ with $\pi^{V\Theta}(v, \theta) > 0$, and every measurable rectangle

$$C = \prod_{i=1}^N C_i \subseteq \prod_{i=1}^N S_i,$$

one has

$$\pi(C \mid v, \theta) = \prod_{i=1}^N \rho(C_i \mid v_i, v_{-i}, \theta).$$

Moreover, the common kernel ρ is invariant to relabelings of opponents: for every bidder i , every own retained value $v_i \in V$, every opponents' retained-value profile $v_{-i} \in V^{N-1}$, every public state $\theta \in \Theta$ with $\pi^{V\Theta}(v_i, v_{-i}, \theta) > 0$, every permutation ψ_{-i} of bidder i 's opponents, and every measurable set $C_i \subseteq S_i$,

$$\rho(C_i \mid v_i, v_{-i}, \theta) = \rho(\psi_{-i}(C_i) \mid v_i, \psi_{-i}(v_{-i}), \theta).$$

This maintained class nests several benchmark cases. Independent private information is obtained when the additional component is degenerate, complete information when it reveals opponents' retained values, and intermediate cases when bidders observe noisy signals about opponents' retained values. The precision of those signals may depend on a bidder's own retained value.

2.4 Symmetric Bayes Nash equilibrium and induced outcome distributions

In an original auction environment, bidder i observes

$$(v_i, s_i, \theta) \in V \times S_i \times \Theta$$

before choosing a bid. A mixed strategy for bidder i is a measurable mapping

$$\sigma_i : V \times S_i \times \Theta \rightarrow \Delta(B),$$

and a strategy profile is denoted by

$$\sigma = (\sigma_1, \dots, \sigma_N).$$

Given $(v, s, \theta) \in V^N \times S \times \Theta$, the induced conditional distribution over bid profiles is

$$\sigma(b \mid v, s, \theta) := \prod_{i=1}^N \sigma_i(b_i \mid v_i, s_i, \theta), \quad b = (b_1, \dots, b_N) \in B^N.$$

We now separate the payoff-relevant auction environment from the latent additional-information structure. Let

$$\mathcal{E}^a := (G^a, \pi^{V\Theta}),$$

where

$$G^a = ((u_i)_{i=1}^N, B^N, \Theta)$$

collects the payoff-relevant auction primitives, and

$$\pi^{V\Theta} \in \Delta(V^N \times \Theta)$$

is the marginal distribution of retained values and public states. Thus $\mathcal{E}^a$ is the given payoff-relevant auction environment; the latent part is the additional-information layer.

Let $\mathcal{I}(\mathcal{E}^a)$ denote the class of admissible original auction information structures consistent with $\mathcal{E}^a$ and satisfying the informational restrictions in Section 2.3. A generic element

$$I \in \mathcal{I}(\mathcal{E}^a)$$

specifies measurable additional-information spaces $(S_i)_{i=1}^N$ and a joint distribution

$$\pi \in \Delta(V^N \times S \times \Theta)$$

whose marginal on $V^N \times \Theta$ is $\pi^{V\Theta}$, and whose latent additional-information layer satisfies the maintained symmetric conditional-independence restriction on the support of $\pi^{V\Theta}$.

A symmetric strategy profile σ is a *symmetric Bayes Nash equilibrium* under $I \in \mathcal{I}(\mathcal{E}^a)$ if no bidder can profitably deviate from it in the original auction environment. For each

$I \in \mathcal{I}(\mathcal{E}^a)$, let $\Sigma^{\text{SBNE}}(G^a, I)$ denote the set of symmetric Bayes Nash equilibrium strategy profiles under I .

Given $\mathcal{E}^a$, an admissible information structure $I \in \mathcal{I}(\mathcal{E}^a)$, and $\sigma \in \Sigma^{\text{SBNE}}(G^a, I)$, define the induced joint distribution over bid profiles, retained-value profiles, and states by

$$\mu_{\sigma, I}(b, v, \theta) := \int_{s \in S} \sigma(b \mid v, s, \theta) \pi(ds, v, \theta), \quad (b, v, \theta) \in B^N \times V^N \times \Theta.$$

Then

$$\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a) := \bigcup_{I \in \mathcal{I}(\mathcal{E}^a)} \{ \mu_{\sigma, I} \in \Delta(B^N \times V^N \times \Theta) : \sigma \in \Sigma^{\text{SBNE}}(G^a, I) \}.$$

This is the set of distributions induced by symmetric Bayes Nash equilibrium as one ranges over admissible symmetric additional-information expansions of $\mathcal{E}^a$.

Under the maintained symmetry assumptions, every distribution $\mu \in \mathcal{M}^{\text{SBNE}}(\mathcal{E}^a)$ satisfies

$$\mu(\varphi(b), \varphi(v), \theta) = \mu(b, v, \theta) \quad \text{for all } (b, v, \theta) \in B^N \times V^N \times \Theta$$

and every permutation φ of $\{1, \dots, N\}$.

The next section introduces the reduced object that characterizes this set of induced distributions within the maintained symmetric auction class.

3 A symmetric reduced object and an equivalence theorem

This section introduces the reduced equilibrium object associated with the maintained symmetric auction class. The object is formulated directly on the given payoff-relevant auction environment $\mathcal{E}^a$, but it preserves the primitive restrictions on symmetric behavior and on the latent additional-information layer. The primitive variable is a symmetric single-bidder recommendation rule over bids, conditional on the bidder's own retained value, an anonymous summary of opponents' retained values, and the public state.

3.1 Anonymous summaries and symmetric recommendation rules

For each bidder i and each opponents' retained-value profile $v_{-i} \in V^{N-1}$, let

$$n_{-i}(v_{-i}) \in \mathbb{Z}_+^{|V|}$$

denote the vector of opponents' retained-value counts, with $\bar{v}$ -th component

$$n_{-i}^{\bar{v}}(v_{-i}) := \sum_{j \neq i} \mathbf{1}\{v_j = \bar{v}\}, \quad \bar{v} \in V.$$

Thus $n_{-i}(v_{-i})$ summarizes opponents' retained values anonymously. Let

$$\mathcal{H} := \left\{ m \in \mathbb{Z}_+^{|V|} : \sum_{\bar{v} \in V} m^{\bar{v}} = N - 1 \right\}$$

denote the set of feasible opponents' count vectors.

A *symmetric reduced recommendation rule* is a stochastic kernel

$$\lambda(\cdot \mid v_i, m, \theta) \in \Delta(B), \quad (v_i, m, \theta) \in V \times \mathcal{H} \times \Theta.$$

The same kernel λ is used for every bidder. Given (v, θ) , the rule induces the conditional distribution over bid profiles

$$\lambda(b \mid v, \theta) = \prod_{i=1}^N \lambda(b_i \mid v_i, n_{-i}(v_{-i}), \theta), \quad b \in B^N.$$

For the given payoff-relevant auction environment $\mathcal{E}^a = (G^a, \pi^{V\Theta})$, the induced distribution over $B^N \times V^N \times \Theta$ is

$$\mu_{\lambda, \mathcal{E}^a}(b, v, \theta) := \pi^{V\Theta}(v, \theta) \prod_{i=1}^N \lambda(b_i \mid v_i, n_{-i}(v_{-i}), \theta), \quad (b, v, \theta) \in B^N \times V^N \times \Theta.$$

Thus the full distribution over bid profiles, retained-value profiles, and the public state is not chosen directly. It is generated from the lower-dimensional symmetric rule λ through the product structure of the reduced recommendation rule.

3.2 Obedience of the reduced object

The recommendation rule can be interpreted through an omniscient mediator. After observing (v, θ) , the mediator draws a bid profile $b \in B^N$ according to the distribution induced by λ and privately recommends b_i to bidder i . The rule may depend on bidder i 's own retained value, the anonymous summary of opponents' retained values, and the public state. Bidder i , however, observes only her own retained value v_i , the public state θ , and the recommended bid b_i . Accordingly, obedience is defined conditional on (v_i, b_i, θ) .

A symmetric reduced recommendation rule λ is *obedient* for $\mathcal{E}^a$ if, for every bidder i , every own retained value $v_i \in V$, every recommended bid $b_i \in B$, every public state $\theta \in \Theta$, and every deviation bid $b'_i \in B$ such that

$$\sum_{v_{-i} \in V^{N-1}} \sum_{b_{-i} \in B^{N-1}} \mu_{\lambda, \mathcal{E}^a}(b_i, b_{-i}, v_i, v_{-i}, \theta) > 0,$$

one has

$$\sum_{v_{-i} \in V^{N-1}} \sum_{b_{-i} \in B^{N-1}} \mu_{\lambda, \mathcal{E}^a}(b_i, b_{-i}, v_i, v_{-i}, \theta) \left[u_i(b_i, b_{-i}, v, \theta) - u_i(b'_i, b_{-i}, v, \theta) \right] \geq 0,$$

where $v = (v_i, v_{-i})$.

Let $\mathcal{R}^{\text{SR}}(\mathcal{E}^a)$ denote the set of obedient symmetric reduced recommendation rules for $\mathcal{E}^a$. Define

$$\mathcal{M}^{\text{SR}}(\mathcal{E}^a) := \{ \mu_{\lambda, \mathcal{E}^a} \in \Delta(B^N \times V^N \times \Theta) : \lambda \in \mathcal{R}^{\text{SR}}(\mathcal{E}^a) \}.$$

This is the set of distributions induced by the reduced symmetric object for the given payoff-relevant auction environment $\mathcal{E}^a$.

3.3 A preliminary equivalence theorem

Theorem 1 (Preliminary symmetric reduction for auctions). *Fix a payoff-relevant auction environment*

$$\mathcal{E}^a = (G^a, \pi^{V\Theta}).$$

Suppose that the payoff primitives G^a satisfy the payoff-symmetry condition in Section 2.2, and that the marginal $\pi^{V\Theta}$ over retained values and public states is exchangeable. Let $\mathcal{I}(\mathcal{E}^a)$ be the class of admissible additional-information expansions whose latent additional-information layer satisfies the maintained symmetric conditional-independence restriction in Section 2.3. Then

$$\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a) = \mathcal{M}^{\text{SR}}(\mathcal{E}^a).$$

That is, a joint distribution over bid profiles, retained-value profiles, and the public state belongs to $\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a)$ if and only if it belongs to $\mathcal{M}^{\text{SR}}(\mathcal{E}^a)$. Equivalently, for a given payoff-relevant auction environment, the distributions generated by symmetric Bayes Nash equilibrium over admissible symmetric additional-information expansions are exactly the distributions generated by the reduced symmetric recommendation rules.

Theorem 1 is a theorem-level formulation rather than a final econometric result. Its role is to identify the reduced object associated with the maintained symmetric auction class and to show that, at the level of induced distributions over (b, v, θ) , no information is lost by working with this reduced object.

The implication for auction inference is that, within the maintained class, one can formulate robust prediction directly in terms of the reduced symmetric recommendation

rule. This is not a claim that the reduced object coincides with unrestricted BCE. Rather, it is the exact reduced-form object generated by the maintained symmetric latent-information class. The next section explains why the resulting characterization does not automatically inherit the linear-programming form of unrestricted BCE.

4 Proof sketch and relation to the main SBCE theorem

This section explains the logic behind Theorem 1. The argument is the auction analogue of the exact-reduction proof in Lu (2026). It uses the same structural ingredients: symmetric payoffs, exchangeability of the retained payoff-relevant component, and symmetric conditional independence of the latent additional-information layer on the support of $\pi^{V\Theta}$.

Fix a payoff-relevant auction environment

$$\mathcal{E}^a = (G^a, \pi^{V\Theta}).$$

The theorem asserts that

$$\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a) = \mathcal{M}^{\text{SR}}(\mathcal{E}^a).$$

It is enough to prove the two inclusions.

4.1 From $\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a)$ to $\mathcal{M}^{\text{SR}}(\mathcal{E}^a)$

Fix

$$\mu_{\sigma, I} \in \mathcal{M}^{\text{SBNE}}(\mathcal{E}^a).$$

By definition, there exist an admissible information structure

$$I \in \mathcal{I}(\mathcal{E}^a)$$

and a symmetric Bayes Nash equilibrium $\sigma \in \Sigma^{\text{SBNE}}(G^a, I)$ such that

$$\mu_{\sigma, I}(b, v, \theta) = \int_{s \in S} \sigma(b \mid v, s, \theta) \pi(ds, v, \theta).$$

Because I is admissible, the conditional law of s given (v, θ) factorizes on the support of $\pi^{V\Theta}$ through a common opponent-symmetric kernel ρ .

Define a symmetric reduced recommendation rule as follows. For each $(v_i, m, \theta) \in V \times \mathcal{H} \times \Theta$, choose any $v_{-i} \in V^{N-1}$ such that $n_{-i}(v_{-i}) = m$ and $\pi^{V\Theta}(v_i, v_{-i}, \theta) > 0$,

whenever such a profile exists, and set

$$\lambda(b_i \mid v_i, m, \theta) := \int_{S_i} \sigma_i(b_i \mid v_i, s_i, \theta) \rho(ds_i \mid v_i, v_{-i}, \theta).$$

If no such v_{-i} exists, define $\lambda(\cdot \mid v_i, m, \theta)$ arbitrarily. This off-support definition does not affect the induced distribution or the obedience inequalities. The symmetry of σ and the opponent-symmetry of ρ imply that the right-hand side depends on v_{-i} only through the anonymous count vector $m = n_{-i}(v_{-i})$, so λ is well defined.

For every (v, θ) with $\pi^{V\Theta}(v, \theta) > 0$, conditional independence gives

$$\mu_{\sigma, I}(b, v, \theta) = \pi^{V\Theta}(v, \theta) \prod_{i=1}^N \lambda(b_i \mid v_i, n_{-i}(v_{-i}), \theta) = \mu_{\lambda, \mathcal{E}^a}(b, v, \theta).$$

Thus the distribution induced by the original equilibrium is generated by the reduced rule λ .

It remains to verify obedience of λ . The obedience inequality for the reduced rule is obtained by averaging the original no-deviation inequalities over bidder i 's latent additional information and over the equilibrium randomization that generates the recommended bid. Any deviation from a recommended bid b_i to another bid b'_i in the reduced problem corresponds to a feasible deviation in the original auction environment that, after the equilibrium draw would have generated b_i , instead chooses b'_i . Since σ is a Bayes Nash equilibrium under I , no such deviation is profitable. Hence λ is obedient for $\mathcal{E}^a$, and

$$\mu_{\sigma, I} = \mu_{\lambda, \mathcal{E}^a} \in \mathcal{M}^{\text{SR}}(\mathcal{E}^a).$$

Therefore

$$\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a) \subseteq \mathcal{M}^{\text{SR}}(\mathcal{E}^a).$$

4.2 From $\mathcal{M}^{\text{SR}}(\mathcal{E}^a)$ to $\mathcal{M}^{\text{SBNE}}(\mathcal{E}^a)$

Conversely, fix

$$\mu_{\lambda, \mathcal{E}^a} \in \mathcal{M}^{\text{SR}}(\mathcal{E}^a).$$

By definition, there exists an obedient symmetric reduced recommendation rule $\lambda \in \mathcal{R}^{\text{SR}}(\mathcal{E}^a)$ such that

$$\mu_{\lambda, \mathcal{E}^a}(b, v, \theta) = \pi^{V\Theta}(v, \theta) \prod_{i=1}^N \lambda(b_i \mid v_i, n_{-i}(v_{-i}), \theta).$$

Construct an admissible original information structure I^λ as follows. Let bidder i 's additional-information space be $S_i = B$, with opponent relabelings acting trivially on S_i . Conditional on (v, θ) , draw the additional-information components independently across bidders according to

$$\rho^\lambda(s_i \mid v_i, v_{-i}, \theta) := \lambda(s_i \mid v_i, n_{-i}(v_{-i}), \theta), \quad s_i \in B.$$

Equivalently, the signal s_i is the bid recommendation for bidder i . The construction is conditionally independent across bidders given (v, θ) , and it is symmetric because the kernel depends on opponents' retained values only through the anonymous count vector. Hence

$$I^\lambda \in \mathcal{I}(\mathcal{E}^a).$$

Define a strategy profile in the reconstructed original auction environment by following the embedded recommendation:

$$\sigma_i^\lambda(b_i \mid v_i, s_i, \theta) = \mathbf{1}\{b_i = s_i\}.$$

This strategy profile is symmetric. Since λ is obedient for $\mathcal{E}^a$, following the recommendation is optimal conditional on (v_i, s_i, θ) . Any mixed deviation is a mixture over deterministic bid deviations, so the obedience inequalities rule out profitable deviations. Thus σ^λ is a symmetric Bayes Nash equilibrium under I^λ .

The induced distribution is exactly

$$\mu_{\sigma^\lambda, I^\lambda} = \mu_{\lambda, \mathcal{E}^a}.$$

Therefore

$$\mathcal{M}^{\text{SR}}(\mathcal{E}^a) \subseteq \mathcal{M}^{\text{SBNE}}(\mathcal{E}^a).$$

Combining the two inclusions yields Theorem 1.

4.3 Relation to the main SBCE theorem

The proof sketch is included to make the auction adaptation transparent. It is a specialization of the general exact-reduction logic in Lu (2026) to a symmetric auction setting with retained payoff-relevant values and latent additional information. The role of the sketch is to identify how the reduced single-bidder recommendation rule arises from the maintained

symmetric latent-information class, and how an obedient reduced rule can be implemented by an admissible original information expansion.

For the purposes of this note, the main unresolved issue is not the existence of an exact reduced object within the maintained symmetric class. The remaining question is how this object should be characterized computationally. That issue is the subject of the next section.

5 Primitive characterization and computational structure

This section explains why the exact symmetric reduction identified above does not, by itself, deliver a linear-programming characterization analogous to the unrestricted BCE characterization used in robust auction inference.

5.1 Unrestricted BCE as a linear feasibility problem

Fix a payoff-relevant auction environment

$$\mathcal{E}^a = (G^a, \pi^{V\Theta}).$$

Under unrestricted BCE, the primitive object is a joint distribution

$$\mu \in \Delta(B^N \times V^N \times \Theta).$$

Bayes consistency requires

$$\sum_{b \in B^N} \mu(b, v, \theta) = \pi^{V\Theta}(v, \theta) \quad \forall (v, \theta) \in V^N \times \Theta.$$

Obedience requires that, for every bidder i , every retained value $v_i \in V$, every recommended bid $b_i \in B$, every public state $\theta \in \Theta$, and every deviation bid $b'_i \in B$,

$$\sum_{v_{-i} \in V^{N-1}} \sum_{b_{-i} \in B^{N-1}} \mu(b_i, b_{-i}, v_i, v_{-i}, \theta) \left[u_i(b_i, b_{-i}, v_i, v_{-i}, \theta) - u_i(b'_i, b_{-i}, v_i, v_{-i}, \theta) \right] \geq 0.$$

This unnormalized inequality is vacuous when the conditioning event (v_i, b_i, θ) has zero probability.

Both Bayes consistency and obedience are linear in the primitive variable μ . Thus, for a given $\mathcal{E}^a$, unrestricted BCE admits a linear feasibility characterization over the full joint distribution. Let

$$\mathcal{M}^{\text{BCE}}(\mathcal{E}^a)$$

denote the set of distributions satisfying these Bayes-consistency and obedience constraints.

5.2 Primitive symmetric formulation

Under the symmetric reduction developed in this note, the primitive object is not a full joint distribution. It is a symmetric single-bidder recommendation rule

$$\lambda(\cdot \mid v_i, m, \theta) \in \Delta(B), \quad (v_i, m, \theta) \in V \times \mathcal{H} \times \Theta,$$

where $\mathcal{H}$ is the set of feasible anonymous opponents' retained-value count vectors defined in Section 3.1. For a retained-value profile $v = (v_i, v_{-i})$, the relevant anonymous summary for bidder i is $m = n_{-i}(v_{-i})$.

Given λ , the induced distribution over bids, retained values, and public states is

$$\mu_{\lambda, \mathcal{E}^a}(b, v, \theta) = \pi^{V\Theta}(v, \theta) \prod_{i=1}^N \lambda(b_i \mid v_i, n_{-i}(v_{-i}), \theta).$$

Thus the full joint distribution over $B^N \times V^N \times \Theta$ is not chosen directly. It is generated from the lower-dimensional rule λ through the product structure imposed by the maintained conditional-independence restriction.

5.3 Polynomial constraints

Bayes consistency is built into the construction of $\mu_{\lambda, \mathcal{E}^a}$:

$$\sum_{b \in B^N} \mu_{\lambda, \mathcal{E}^a}(b, v, \theta) = \pi^{V\Theta}(v, \theta) \quad \forall (v, \theta) \in V^N \times \Theta.$$

The substantive feasibility restriction is obedience. Substituting the induced distribution into the reduced obedience inequalities yields, for every bidder i , every $v_i \in V$, every recommended bid $b_i \in B$, every public state $\theta \in \Theta$, and every deviation $b'_i \in B$,

$$\sum_{v_{-i} \in V^{N-1}} \sum_{b_{-i} \in B^{N-1}} \pi^{V\Theta}(v, \theta) \prod_{j=1}^N \lambda(b_j \mid v_j, n_{-j}(v_{-j}), \theta) \left[u_i(b_i, b_{-i}, v_i, v_{-i}, \theta) - u_i(b'_i, b_{-i}, v_i, v_{-i}, \theta) \right] \geq 0,$$

where inside the summation $v = (v_i, v_{-i})$ and $b = (b_i, b_{-i})$. These inequalities are no longer linear in the primitive variables $\lambda(b_i \mid v_i, m, \theta)$. The product term

$$\prod_{j=1}^N \lambda(b_j \mid v_j, n_{-j}(v_{-j}), \theta)$$

introduces polynomial structure. Hence the primitive symmetric formulation leads naturally to a polynomial feasibility problem.

In addition, λ must satisfy the simplex constraints

$$\lambda(b_i \mid v_i, m, \theta) \geq 0 \quad \forall b_i \in B,$$

and

$$\sum_{b_i \in B} \lambda(b_i \mid v_i, m, \theta) = 1$$

for every $(v_i, m, \theta) \in V \times \mathcal{H} \times \Theta$. Accordingly, the primitive symmetric characterization is a feasibility problem in the variables

$$\{\lambda(b_i \mid v_i, m, \theta) : (b_i, v_i, m, \theta) \in B \times V \times \mathcal{H} \times \Theta\},$$

subject to simplex constraints and polynomial obedience inequalities.

5.4 The relevant tradeoff

The comparison with unrestricted BCE is therefore not simply linear versus nonlinear. The relevant tradeoff is between full-joint unrestricted search and lower-dimensional structured search.

Under unrestricted BCE, the analyst searches over

$$\mu \in \Delta(B^N \times V^N \times \Theta),$$

and the constraints are linear in μ . This object is flexible, but its dimension grows quickly with the number of bidders, bid levels, retained values, and public states.

Under the primitive symmetric formulation, the analyst searches over the smaller single-bidder rule

$$\lambda(\cdot \mid v_i, m, \theta), \quad (v_i, m, \theta) \in V \times \mathcal{H} \times \Theta.$$

Up to simplex normalizations, the number of primitive probability vectors is $|V| |\mathcal{H}| |\Theta|$, where

$$|\mathcal{H}| = \binom{N + |V| - 2}{|V| - 1}.$$

The reduction is therefore potentially substantial, especially when the full joint bid-value space is large. The cost is that feasibility is no longer a linear program in the primitive variables.

Thus the theorem-level reduction delivers an exact lower-dimensional primitive object for the maintained symmetric class, but it does not automatically preserve the linear-programming form of unrestricted BCE. For the purposes of this note, this observation is the main computational implication. A full analysis of how best to solve, lift, relax, or approximate the resulting polynomial feasibility problem is left for future work.

6 Scope and open questions

This note is a theorem-oriented preliminary extension of the SBCE agenda to symmetric auction environments with latent information. For a given payoff-relevant auction environment, it identifies a maintained symmetric latent-information class, defines the corresponding reduced symmetric object, and records a theorem-level exact reduction result. The note does not provide a finished econometric implementation or a final computational characterization.

The scope is deliberately limited. The reduced object is not proposed as a general-purpose substitute for unrestricted BCE. Unrestricted BCE remains the natural robust object when the analyst does not maintain the symmetric latent-information class studied here. The point of the note is instead to ask what the appropriate reduced-form object is once that primitive class is maintained. Within the class, the answer is the reduced symmetric recommendation rule characterized above.

Several questions remain open. First, the primitive-search formulation in Section 5 may admit a lifted representation that restores linearity or convexity without sacrificing exactness. Whether such a representation exists, and whether it is computationally useful, remains to be worked out.

Second, the maintained symmetric class may have implications beyond dimensional reduction. In some auction environments, the restrictions on symmetric behavior and on the latent additional-information layer may sharpen counterfactual or identified sets relative to unrestricted BCE. Determining when this happens requires concrete auction applications.

Third, a full development should study benchmark formats in which the computational tradeoff can be evaluated directly. Natural starting points include symmetric first-price auctions with discretized private values and common-value auctions with finitely supported

retained payoff-relevant states. Such examples would allow one to compare the full-joint unrestricted BCE formulation with the lower-dimensional primitive symmetric formulation in repeated feasibility or inference exercises.

The contribution of this note is therefore limited but clear. It identifies the auction-facing reduced object associated with a maintained symmetric latent-information class, clarifies how that object relates to the unrestricted BCE framework of Syrgkanis et al. (2021), and isolates the computational issue created by the move from full-joint unrestricted search to exact lower-dimensional structured search.

References

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- Syrgkanis, V., E. Tamer, and J. Ziani (2021). Inference on auctions with weak assumptions on information. Manuscript, Harvard University, Department of Economics.